

Marshall Asch

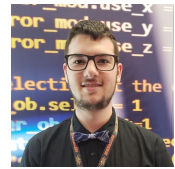
✉ hello@marshallasch.ca

🐦 @MarshallAsch

🌐 @MarshallAsch

🌐 marshallasch.ca

in in/MarshallAsch



I am a Senior DevOps Engineer at Plex, where I help run a multi-cloud (AWS and GCP) Kubernetes platform using Terraform, ArgoCD GitOps, and OpenTelemetry-based observability. Over the past several years I have led cluster migrations, a production ingress migration to Envoy Gateway, multi-cluster cost attribution, and a steady stream of security and cost clean-ups.

I have a passion for command-line tools, networking, servers, and open source projects. Outside of work I build and self-host full stack applications, including a multi-tenant SaaS platform for dive shops, on infrastructure I manage end to end.

Education

- 2019 – 2022 **M.Sc. Computer Science, University of Guelph, Canada** in Mesh Networking.
Thesis: *Replicability and Efficacy of the 'Replication in Highly Partitioned Mobile ad hoc Networks' Data Replication Scheme* 📄
- 2015 – 2019 **B.Comp. (Hons) Computer Science, University of Guelph, Canada.**
Degree with Distinction Recognition for Outstanding Academic Performance.

Projects

- 03/2026 - Present **dive-tec** 🌐 Multi-tenant SaaS platform for dive shops: each shop gets course scheduling, e-commerce, trip and event booking, equipment rentals, and staff management under its own brand and domain.
- 11/2025 - Present **Fill Station** 🌐 Open source (AGPL-3.0) web app for tracking scuba cylinder fills, visual inspections, and compressor maintenance at a dive shop.
- 10/2021 - 2024 **OO Resources** 🌐 Offline Open source teaching resource to provide interactive course content and development dependencies in one place that does not require internet access.
- 05/2021 - 2022 **ns-3 Github CI pipeline** 🌐 Continuous development tool to verify that modules built of the ns-3 network simulator compile and run as intended.
- 2021 - 2022 **ns-3 Static Access Frequency Module** 🌐 Implemented and maintained a module for the ns-3 network simulator.
- ns-3 RHPMAN Module** 🌐 Implemented and maintained a module for the ns-3 network simulator.
- 2020 - 2024 **liquidctl maintainer** 🌐 Maintainer on the python liquid cooler controller CLI application
- 2021 - 2025 **Botomir maintainer** 🌐 Maintainer on the nodeJS Discord Bot application

Certification and Training

- 07/2026 **Normoxic Trimix Diver.** Technical Diving International (TDI).
- Normoxic Trimix Diver.** IANTD.
- 10/2025 **Trimix Gas Blender.** IANTD.
- Advanced Gas Blender.** Scuba Diving International (SDI).
- 03/2025 **Divemaster.** Professional Association of Diving Instructors (PADI). Cert. #564289.
- 01/2025 **Nitrox Gas Blender.** Scuba Diving International (SDI).
- Cylinder O₂ Service Technician.** Scuba Diving International (SDI).
- Visual Inspection Technician.** IANTD. (expires 01/2028)

Projects (continued)

- 07/2024
 - **VIP Inspector.** Scuba Diving International (SDI). (expires 01/2028)
 - **Air Fill Station Technician.** Scuba Diving International (SDI).
 - **Decompression Procedures.** Scuba Diving International (SDI).
 - **Advanced Nitrox.** Scuba Diving International (SDI).
- 03/2024
 - **Oxygen Administration.** Divers Alert Network (DAN).
- 2021
 - **M001 - MongoDB Basics Course.** MonogoDB University.
 - **M103 - MongoDB Basic Cluster Administration Course.** MonogoDB University.
 - **M121 - MongoDB Aggregation Framework Course.** MonogoDB University.

Employment History

- 09/2026 – Present
 - **Sr. DevOps Engineer**, Plex, Inc., Remote.
Promoted from DevOps Engineer. Technical owner for the multi-cloud (AWS + GCP) Kubernetes platform, Terraform IaC, and GitOps delivery.
- 05/2022 – 09/2026
 - 2025 – 2026
 - **DevOps Engineer**, Plex, Inc., Remote.
 - **Divemaster** (part-time), Scuba 2000, Toronto, ON.
Assisted instructors on Open Water scuba courses, classroom and pool intro-to-scuba sessions, and specialty courses.
- 01/2021 – 05/2022
 - **DevOps Engineer**, Value Connect, Guelph, ON.
Promoted from Jr. Software Developer (01/2021) to Jr. DevOps Engineer (09/2021) to DevOps Engineer (01/2022).
- 01/2021 – 06/2021
 - **Graduate Teaching Assistant (Compilers)**, University of Guelph, ON.
- 09/2020 – 12/2020
 - **Graduate Teaching Assistant (Cloud Computing)**, University of Guelph, ON.
- 01/2020 – 04/2020
 - **Graduate Teaching Assistant (Operating Systems)**, University of Guelph, ON.
- 09/2019 – 08/2020
 - **Left Mitacs Accelerate Intern**, University of Guelph, ON.
- 09/2019 – 12/2019
 - **Graduate Teaching Assistant (Microcomputers)**, University of Guelph, ON.
- 05/2019 – 08/2019
 - **University of Guelph Mitacs Accelerate Intern**, Left, Maple Ridge, BC.
- 01/2019 – 04/2019
 - **Lead Developer and Undergraduate Research Assistant**, Working on the eNuk project, University of Guelph, ON.
 - **Undergraduate Teaching Assistant (Introduction to Computer Applications)**, University of Guelph, ON.
- 05/2018 – 08/2018
 - **Mobile App Developer**, Left, Maple Ridge, BC.
- 05/2017 – 08/2017
 - **IT Student**, Lakeridge Health, Oshawa, ON.
- 06/2016 – 08/2016
 - **High Ropes Instructor**, Camp New Moon, ON
- 06/2015 – 08/2015
 - **High Ropes Instructor**, Camp New Moon, ON
- 06/2014 – 08/2014
 - **High Ropes Instructor**, Camp New Moon, ON

Volunteer and Extra-Curricular Experience

Volunteering

- 01/2021 - Present
 - **Road2IR**, Provided technical and networking infrastructure advice.
- 10/2019
 - **Ontario University Fair**, Represented the University of Guelph, Toronto, ON.
- 09/2019 – 04/2020
 - **Guelph Coding Community**, Student club executive, Guelph, ON.
- 09/2019 – 12/2019
 - **Climbing Gym Monitor**, University of Guelph, ON.

Volunteer and Extra-Curricular Experience (continued)

- 12/2019 ■ **Major League Hacking, Local Hack Day event Organizer**, Society of Computing and Information Science, Guelph, ON.
- 12/2018 ■ **Major League Hacking, Local Hack Day event Organizer**, Society of Computing and Information Science, Guelph, ON.
- 09/2018 – 04/2020 ■ **Society of Computing and Information Science, System Administrator**, University of Guelph, ON.
- 04/2017 – 06/2018 ■ **Full Stack Developer**, ActivNtel, Guelph, ON.
- 01/2016 – 02/2017 ■ **Climbing Gym Monitor**, University of Guelph, ON.
- 09/2016 – 06/2017 ■ **Web Developer**, NGTronix, Guelph, ON.

Committees

- 09/2019 – 03/2022 ■ **IT Strategy Advisory Committee**. University of Guelph, ON.
- 01/2018 – 04/2019 ■ **Roboticon Planning Committee** Society of Computing and Information Science, Guelph, ON.
- 09/2017 – 04/2019 ■ **Computer Infrastructure Committee** School of Computer Science, University of Guelph, ON.
- 09/2018 – 04/2019 ■ **Curriculum Committee** School of Computer Science, University of Guelph, ON.

Awards and Recognitions

- 09/2020 ■ **University of Guelph GSA Conference Runner up**, Awarded to the two runner up presenters for the student run conference.
- 04/2020 ■ **Erin Angus Graduate Student Award**. Awarded to a graduate student for making significant contributions to student life by Graduate Student Association.
- **CEPS Dean's Tri-Council Scholarship**, (\$5000) Deans Scholarship awarded to NSERC award holders, University of Guelph, ON.
- **Canada Graduate Scholarships-Master's Program**, (\$17500) Natural Sciences and Engineering Research Council of Canada (NSERC) Scholarship, University of Guelph, ON.
- 02/2020 ■ **Ontario Graduate Scholarship**, (\$5000) A merit based academic scholarship, University of Guelph, ON.
- 05/2019 ■ **College of Engineering and Physical Sciences Society of Excellence**, Demonstrated academic excellence and outstanding contributions to the university community, University of Guelph, ON.
- **College of Engineering and Physical Science Deans Honours List**, University of Guelph, ON. F15, W16, F16, W17, F17, W18, F18, W19
- 10/2018 ■ **Dr. James Linders Scholarship**. (\$500) Awarded for achieving the highest third year average by the School of Computer Science, University of Guelph, ON.
- 09/2015 ■ **University of Guelph Entrance Scholarship**. (\$2000) Awarded for being admitted with an average greater than 85% by the University of Guelph, ON.

Research Grants

- 04/2020 ■ **Northern Scientific Training Program**. (\$2150) Awarded to conduct fieldwork in Northern Canada by Polar Knowledge Canada.
- 04/2019 ■ **Northern Scientific Training Program**. (\$2307) Awarded to conduct fieldwork in Northern Canada by Polar Knowledge Canada.

Non Academic Presentations

- 09/2020 ■ **Introduction to Git**, Guelph Coding Community, Guelph, ON.
- 03/2020 ■ **MSc. Seminar - Incorporation of Human Mobility in Data Replication Schemes in Mobile Ad Hoc Networks**, Guelph Coding Community, Guelph, ON.
- 09/2019 ■ **Advanced Git**, Guelph Coding Community, Guelph, ON.
- **Introduction to BASH and Git**, Guelph Coding Community, Guelph, ON.
- 04/2019 ■ **Bridging the Digital Divide with Mobile Ad Hoc Networks in the Circumpolar North**, Guelph Coding Community, Guelph, ON.
- 12/2018 ■ **Major League Hacking Localhost Workshop**, Society of Computing and Information Science, Guelph, ON.
- 11/2018 ■ **Mesh Networking**, Guelph Coding Community, Guelph, ON.

Research Publications

Journal Articles

- 1 Savić, I., **Asch, M.**, Rourke, K., Safari, F., Houlding, P., Fraeys de Veubeke, J., Ernst, J., & Gillis, D. (2021). M-ODD: A Standard Protocol for Reporting MANET Related Models, Simulations, and Findings [in press].
- 2 **Asch, M.**, Seymour, P. F., Ernst, J., & Gillis, D. (2019). Incorporation of Node Mobility in Data Replication Schemes in Mobile Ad Hoc Networks [Oral presentation and Preceding]. *2019 IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, 0230–0236. <https://doi.org/10.1109/IEMCON.2019.8936268>
- 3 Durish, N., Gillis, D., Seymour, P. F., Rourke, K., **Asch, M.**, Raturi, S., Hughes, B., Flowers, C., Shiwak, I., Wood, M., & Ernst, J. (2019). Bridging the digital divide in the circumpolar north [abstract]. *Labrador Research Forum*.
- 4 Gillis, D., Durish, N., Rourke, K., **Asch, M.**, Knarr, J., Samuels, J., Kipp, A., Sawatzky, A., Middleton, J., Borish, D., Bell, I., Cunsolo, A., and Charlie Flowers, I. S., Wood, M., & Harper, S. L. (2019). Supporting community-based monitoring with technology in the circumpolar north [abstract]. *Labrador Research Forum*.
- 5 Durish, N., Gillis, D., Seymour, P. F., Rourke, K., **Asch, M.**, Raturi, S., Hughes, B., Flowers, C., Shiwak, I., Wood, M., & Ernst, J. (2018). Bridging the digital divide to support community-based monitoring in the circumpolar north [abstract]. *ArcticNet*.

Conference Presentations

- 1 **Asch, M.**, Lin, X., & Gillis, D. (2020, August). Impact of social interactions on mobile ad hoc networks.
- 2 **Asch, M.**, Rourke, K., Seymour, P. F., Durish, N., Flowers, C., Wood, M., Gillis, D., & Ernst, J. (2019, March). Bridging the digital divide with mobile ad hoc networks in the circumpolar north.

Skills

- Languages ■ Go, Python, TypeScript/Javascript, BASH, SQL, Java, C/C++, $\text{\LaTeX}$
- Cloud ■ AWS (EKS, IAM, VPC, NLB, PrivateLink, RDS, S3, Athena), GCP (GKE, Artifact Registry, Cloud SQL), Cloudflare.
- Platform ■ Kubernetes, Helm, ArgoCD, Kustomize, Envoy Gateway, Kyverno, KEDA, External Secrets, cert-manager, Vault.

Skills (continued)

- IaC & CI/CD 📌 Terraform, Atlantis, GitHub Actions, Renovate, Docker, Docker-Compose, Bitbucket Pipelines.
- Observability 📌 OpenTelemetry, SigNoz, Datadog, Prometheus, Grafana, OpenCost.
- Databases 📌 Postgres, MariaDB/MySQL, MongoDB, Valkey/Redis, RabbitMQ.

References

Available on Request

Last updated: 26th September 2026